Online Appendix

Austerity, Democracy, and Coercion: A Model of Fiscal Pressure and the Severity of Protest Policing

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A.1 Policy regions and comparative statics for the additive-quadratic benchmark.

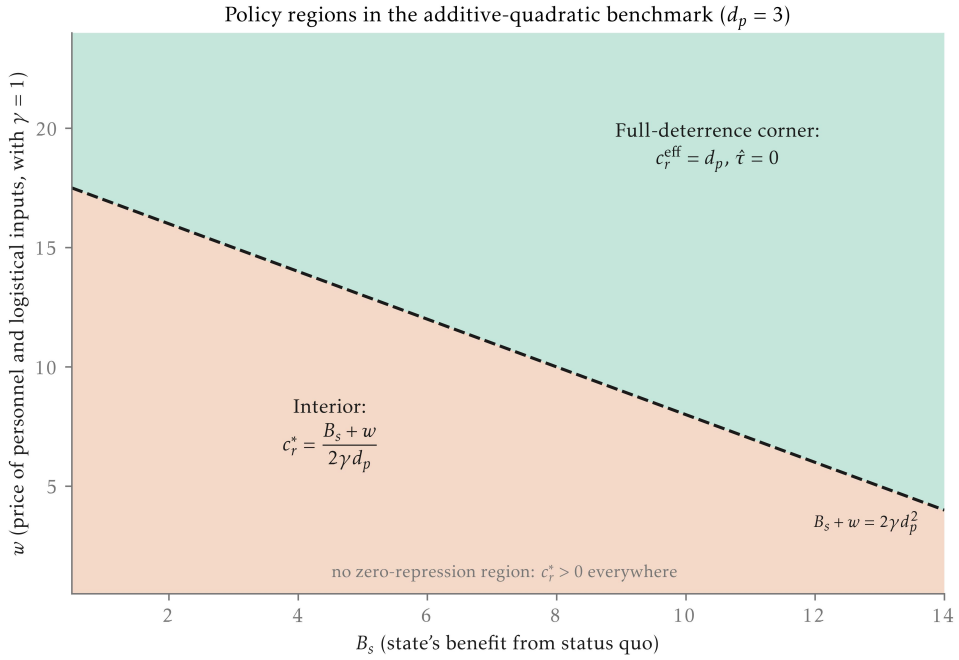


Figure A1: Interior and full-deterrence regions for $V(\bar{N}, c_r; w) = w\bar{N}$ and $P(c_r; \gamma) = \gamma c_r^2$. The boundary is $B_s + w = 2\gamma d_p^2$.

A.2 Posterior beliefs under the common proper prior.

Lemma 1 (Proper-prior posterior)**.** *Suppose* $\tau \sim U[0,1]$ *and* $x_i = \tau + \sigma\varepsilon_i$*, where* ε_i *has density* f *and c.d.f.* F*. For* $t \in [0,1]$*,*

$$\Pr(\tau \le t \mid x_i = x) = \frac{F(x/\sigma) - F((x-t)/\sigma)}{F(x/\sigma) - F((x-1)/\sigma)}.$$

Proof. The likelihood of x conditional on τ is $f((x-\tau)/\sigma)/\sigma$. The uniform prior restricts

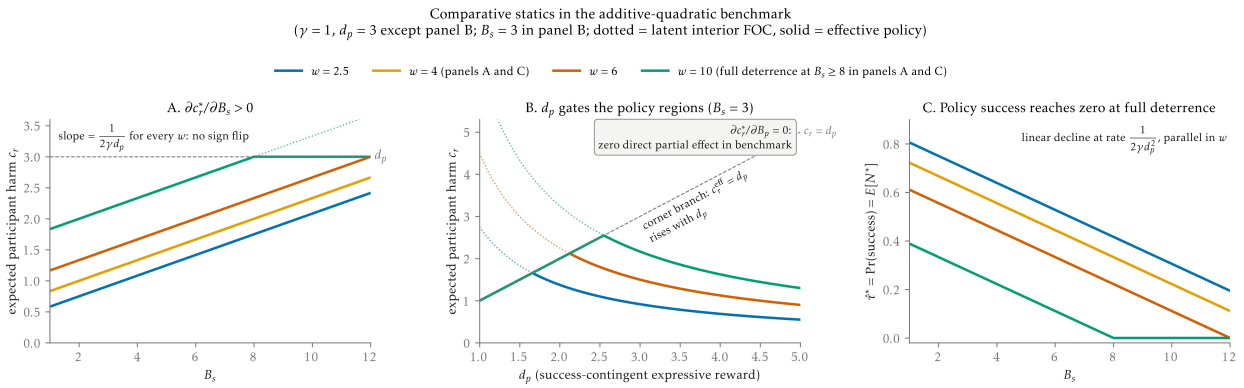


Figure A2: Region-conditional comparative statics for the additive-quadratic benchmark. Panel A varies the government's status quo stake. Panel B varies success-contingent motivation. Panel C shows the corresponding probability of policy change. Dotted curves are unconstrained interior solutions and solid curves are effective policies. The vertical axes label c_r as expected participant harm, not harm conditional on targeting.

its support to $[0,1]$. Substituting $u = (x-\tau)/\sigma$ gives

$$\int_0^t \frac{1}{\sigma} f\left(\frac{x-\tau}{\sigma}\right) d\tau = F(x/\sigma) - F((x-t)/\sigma),$$
$$\int_0^1 \frac{1}{\sigma} f\left(\frac{x-\tau}{\sigma}\right) d\tau = F(x/\sigma) - F((x-1)/\sigma).$$

Their ratio is the posterior probability. □

Lemma 2 (Monotone posterior)**.** *Under the log-concavity assumption on f, the posterior probability $\Pr(\tau \leq t \mid x_i = x)$ is nonincreasing in x for every fixed $t \in (0,1)$. A marginal-indifference solution to equation* (1) *therefore supports a monotone cutoff best response.*

Proof. For $x' > x$, the likelihood ratio of the two signals is

$$\frac{f((x'-\tau)/\sigma)}{f((x-\tau)/\sigma)}.$$

Log-concavity of f makes this ratio nondecreasing in τ: writing $u = (x-\tau)/\sigma$ and $a = (x'-x)/\sigma > 0$, the ratio $f(u+a)/f(u)$ is nonincreasing in u, while u decreases with τ. The posterior after x' therefore dominates the posterior after x in the monotone-likelihood-

ratio order. Higher signals place more probability on higher thresholds, so $\Pr(\tau \le t \mid x_i = x)$ is nonincreasing in x. At a solution to equation (1), citizens below the indifferent signal weakly prefer participation and citizens above it weakly prefer abstention. □

A.3 Proof of Proposition 1.

Proof of Proposition 1. Fix a cutoff strategy profile with cutoff $\hat{x}$: citizen i participates if and only if $x_i < \hat{x}$.

Step 1: participation and the induced success threshold. At true threshold τ, the mass of participants is

$$N^*(\tau) = \Pr(x_i < \hat{x} \mid \tau) = \Pr(\tau + \sigma\varepsilon_i < \hat{x}) = \Pr\left(\varepsilon_i < \frac{\hat{x} - \tau}{\sigma}\right) = F\left(\frac{\hat{x} - \tau}{\sigma}\right),$$

which is continuous and strictly decreasing in τ. At $\tau = 0$, $N^*(0) > 0$, while at $\tau = 1$, $N^*(1) < 1$. Hence $N^*(\tau) - \tau$ crosses zero exactly once on $(0, 1)$. Denote the crossing by $\hat{\tau}_\sigma$. It satisfies

$$F\left(\frac{\hat{x}_\sigma - \hat{\tau}_\sigma}{\sigma}\right) = \hat{\tau}_\sigma, \tag{10}$$

and protest succeeds exactly when $\tau \le \hat{\tau}_\sigma$.

Step 2: reduce the system to one dimension. A citizen with signal x_i gains $\Pr(\tau \le \hat{\tau}_\sigma \mid x_i)d_p - c_r$ from participating. Lemma 1, evaluated at $t = \hat{\tau}_\sigma$ and $x_i = \hat{x}_\sigma$, gives

$$\Pr(\tau \le \hat{\tau}_\sigma \mid x_i = \hat{x}_\sigma) = \frac{F(\hat{x}_\sigma/\sigma) - F((\hat{x}_\sigma - \hat{\tau}_\sigma)/\sigma)}{F(\hat{x}_\sigma/\sigma) - F((\hat{x}_\sigma - 1)/\sigma)}.$$

Together with marginal indifference, this expression and equation (10) give the exact system in equation (1).

For any candidate threshold $t \in (0, 1)$, define $z(t) = F^{-1}(t)$ and

$$x_\sigma(t) = t + \sigma z(t).$$

This pair satisfies the crossing condition because $F((x_\sigma(t)-t)/\sigma) = F(z(t)) = t$. Substitution into the marginal citizen's posterior gives the continuous function

$$\Pi_\sigma(t) \equiv \pi_\sigma(x_\sigma(t), t) = \frac{F(t/\sigma + z(t)) - t}{F(t/\sigma + z(t)) - F((t-1)/\sigma + z(t))}. \tag{11}$$

An interior solution is a root of $\Pi_\sigma(t) = \rho$, where $\rho = c_r/d_p$.

Step 3: local existence and compactness. Let $t_0 = 1 - \rho$ and fix $\delta \in (0, \min\{t_0, 1-t_0\})$. On the compact interval $I_\delta = [t_0 - \delta, t_0 + \delta]$, $z(t)$ is bounded. Uniformly for $t \in I_\delta$,

$$F(t/\sigma + z(t)) \longrightarrow 1, \qquad F((t-1)/\sigma + z(t)) \longrightarrow 0$$

as $\sigma \to 0$. Equation (11) therefore implies the uniform convergence

$$\sup_{t \in I_\delta} |\Pi_\sigma(t) - (1-t)| \longrightarrow 0.$$

At the left endpoint, the limiting value is $1 - (t_0 - \delta) = \rho + \delta$; at the right endpoint it is $1 - (t_0 + \delta) = \rho - \delta$. For all sufficiently small positive σ, continuity then gives

$$\Pi_\sigma(t_0 - \delta) > \rho > \Pi_\sigma(t_0 + \delta).$$

The intermediate value theorem supplies a root $\hat{\tau}_\sigma \in I_\delta$ for every such σ. Lemma 2 makes the associated cutoff $\hat{x}_\sigma = x_\sigma(\hat{\tau}_\sigma)$ a best response, so the root is an interior monotone cutoff equilibrium. Its compactness follows from the construction rather than from a separate assumption.

Step 4: convergence of the selected branch. For any selection of roots inside I_δ,

$$|\hat{\tau}_\sigma - t_0| = |(1 - \hat{\tau}_\sigma) - \rho| \leq \sup_{t \in I_\delta} |\Pi_\sigma(t) - (1-t)| \longrightarrow 0.$$

Because $F^{-1}(\hat{\tau}_\sigma)$ is bounded on I_δ,

$$\hat{x}_\sigma - \hat{\tau}_\sigma = \sigma F^{-1}(\hat{\tau}_\sigma) \longrightarrow 0.$$

Hence both equilibrium objects converge to $t_0 = 1 - c_r/d_p$.

Step 5: boundary policies. If $c_r > d_p$, the participation gain is strictly negative even when success is certain, so abstention is uniquely optimal. At $c_r = d_p$, the gain is weakly negative and equals zero when success is certain. Universal abstention and universal participation can therefore both be supported as weak equilibria. The selected full-deterrence outcome is the continuous extension $\hat{\tau}(c_r) \downarrow 0$ as $c_r \uparrow d_p$. At $c_r = 0$, participation is weakly dominant, and full participation is the corresponding continuation $\hat{\tau}(c_r) \uparrow 1$ as $c_r \downarrow 0$. □

Remark 1. At positive noise, equation (1) can have additional roots near the boundaries of the common prior. Proposition 1 selects the root in a fixed neighborhood of $1 - c_r/d_p$ whose existence is guaranteed for all sufficiently small noise. Every selection from that neighborhood has the same limit. The state's problem is defined using this limiting branch.[17]

For $0 < c_r < d_p$, universal participation and universal abstention are also strict Bayesian equilibria at every positive noise level: the participation advantage is respectively $d_p - c_r > 0$ and $-c_r < 0$. The state results are conditional on the interior limiting selection specified in Proposition 1 and its stated endpoint extensions.

A.4 Local positive-noise robustness.

The state problem above uses the small-noise limit. To assess whether the crowd-cost result persists when signals remain noisy, let Φ denote the standard normal c.d.f. For a candidate threshold $t \in (0,1)$, the first equation in (1) implies $x_\sigma(t) = t + \sigma\Phi^{-1}(t)$. Substituting this

17. The locally diffuse limiting argument follows the standard global-games logic in Morris and Shin (2003) and Bueno de Mesquita and Shadmehr (2023).

expression into the marginal citizen's posterior reduces the two-equation system to the one-dimensional residual in equation (11). When the bounded prior generates more than one root, we follow the interior branch specified in Proposition 1.

For each selected root $\hat{\tau}_\sigma$, expected protest mass is calculated directly as

$$\mathbb{E}_\tau[N^*(\tau)] = \int_0^1 \Phi\left(\frac{x_\sigma(\hat{\tau}_\sigma) - \tau}{\sigma}\right) d\tau,$$

The state payoff uses this expected mass rather than imposing its limiting equality with the success probability. For $B_s = 3$, $d_p = 3$, $\gamma = 1$, $V(\bar{N}, c_r; w) = w\bar{N}$, and $P(c_r; \gamma) = \gamma c_r^2$, the table reports numerical local maximizers of the state payoff along the interior citizen-equilibrium branch. These calculations illustrate the local crowd-cost response; they do not establish global optimality across policies or equilibrium branches.

σ	$c_{r,\sigma}^{\text{loc}}$ at $w = 1.0$	$c_{r,\sigma}^{\text{loc}}$ at $w = 1.1$
0.05	0.6947	0.7134
0.03	0.6832	0.7013
0.02	0.6778	0.6954
0.01	0.6722	0.6894

At every reported noise level, raising w from 1.0 to 1.1 raises the local maximizing policy. Both reported policy sequences move toward their respective analytic limits, 0.6667 and 0.6833, as noise falls. Expected protest mass also approaches the success probability: at $c_r/d_p = 0.2$, the absolute gap falls from 0.0421 at $\sigma = 0.05$ to 0.0084 at $\sigma = 0.01$.

A.5 Proof of Proposition 2.

Proof of Proposition 2. For $c_r \in [0, d_p]$, substitute $\hat{\tau}(c_r) = 1 - c_r/d_p$ into equation (3):

$$u_S(c_r) = \frac{c_r}{d_p} B_s - V\left(1 - \frac{c_r}{d_p}, c_r; w\right) - P(c_r; \gamma).$$

Differentiation gives

$$u_c(c_r) = \frac{B_s + V_N}{d_p} - V_c - P_c,$$
$$u_{cc}(c_r) = -\frac{V_{NN}}{d_p^2} + \frac{2V_{Nc}}{d_p} - V_{cc} - P_{cc}.$$

Assumption 5 makes the objective strictly concave on $(0, d_p)$ and gives $u_c(0^+) > 0$. If $u_c(d_p^-) < 0$, continuity of the interior derivative and strict concavity imply a unique root of $u_c(c_r) = 0$ in $(0, d_p)$, which is the unique maximizer. If $u_c(d_p^-) \geq 0$, the objective is increasing up to the full-deterrence corner and is maximized at $c_r = d_p$ on this interval. At every $c_r > d_p$, the clamped threshold and expected protest mass are zero, so $u_S(c_r) = B_s - V(0, c_r; w) - P(c_r; \gamma)$ and its derivative is $-V_c(0, c_r; w) - P_c(c_r; \gamma) < 0$. Thus the policy characterized in the proposition is globally optimal. This argument uses one-sided derivatives at the kink $c_r = d_p$. □

A.6 Proof of Proposition 3.

Proof of Proposition 3. At an interior solution define[18]

$$G(c_r; B_s, w, \gamma, d_p) \equiv u_c(c_r) = 0.$$

Because $G_{c_r} = u_{cc} < 0$, the implicit function theorem gives

$$\frac{\partial c_r^*}{\partial B_s} = -\frac{1/d_p}{u_{cc}} > 0, \qquad \frac{\partial c_r^*}{\partial \gamma} = \frac{P_{c\gamma}}{u_{cc}} < 0,$$

where $P_{c\gamma} > 0$. Fixing c_r and differentiating G in d_p gives

$$G_{d_p} = -\frac{B_s + V_N + c_r V_{Nc} - (c_r/d_p) V_{NN}}{d_p^2},$$

and therefore

$$\frac{\partial c_r^*}{\partial d_p} = \frac{B_s + V_N + c_r^* V_{Nc} - (c_r^*/d_p) V_{NN}}{d_p^2 u_{cc}}.$$

This derivative is negative under the condition stated in Proposition 3. For the crowd-cost shifter,

$$G_w = \frac{V_{Nw}}{d_p} - V_{cw}, \qquad \frac{\partial c_r^*}{\partial w} = -\frac{V_{Nw}/d_p - V_{cw}}{u_{cc}}.$$

Since $u_{cc} < 0$, this derivative is positive exactly when the numerator is positive. This is the crowd-cost condition. The primitive B_p is absent from G, which gives its zero direct partial effect.

Finally, $\hat{\tau}^* = 1 - c_r^*/d_p$. Its response to B_s, w, and γ has the opposite sign from the

18. For the first-order-condition function G, $G_z \equiv \partial G/\partial z$ fixes every displayed primitive other than z. In particular, $G_{c_r} = u_{cc}$, while G_{d_p} and G_w differentiate the first-order condition at a fixed candidate policy c_r. In the limiting state problem, expected mass equals $\hat{\tau} = 1 - c_r/d_p$, so G_{d_p} includes the induced change $\partial\hat{\tau}/\partial d_p = c_r/d_p^2$ at fixed c_r. The implicit-function step then allows the equilibrium policy to adjust.

corresponding response of c_r^*. For motivation,

$$\frac{\partial \hat{\tau}^*}{\partial d_p} = \frac{c_r^*}{d_p^2} - \frac{1}{d_p}\frac{\partial c_r^*}{\partial d_p} > 0,$$

because $c_r^* > 0$ and $\partial c_r^*/\partial d_p < 0$. The direct partial effect of B_p on $\hat{\tau}^*$ is also zero. □

A.7 Proof of Proposition 4.

Proof of Proposition 4. At an interior threshold, with $d_p(c_r) = d_0 + e(c_r) > 0$, apply the quotient rule to $\hat{\tau}(c_r) = 1 - c_r/d_p(c_r)$:

$$\frac{\partial \hat{\tau}}{\partial c_r} = -\frac{d_p(c_r)\cdot 1 - c_r\, e'(c_r)}{d_p(c_r)^2} = -\frac{d_p(c_r) - c_r\, e'(c_r)}{d_p(c_r)^2}.$$

The denominator is positive, so the derivative has the sign of $-(d_p(c_r) - c_r e'(c_r))$. Dividing the parenthesis by $d_p(c_r) > 0$,

$$d_p(c_r) - c_r\, e'(c_r) \gtrless 0 \quad \Longleftrightarrow \quad 1 - \frac{c_r\, e'(c_r)}{d_p(c_r)} = 1 - \varepsilon(c_r) \gtrless 0,$$

so $\partial \hat{\tau}/\partial c_r < 0$ exactly when $\varepsilon(c_r) < 1$ and $\partial \hat{\tau}/\partial c_r > 0$ exactly when $\varepsilon(c_r) > 1$. □

A.8 Proof of Proposition 5.

Proof of Proposition 5. At an interior threshold, equation (9) gives $D = -\hat{\tau}_c$. Differentiating the state payoff at a fixed α gives the first-order condition

$$u_c(c_r; \alpha) = (B_s + V_N)\, D - V_c - P_c = 0.$$

At a positive policy, $B_s + V_N > 0$, $V_c \geq 0$, and $P_c > 0$. The equality is therefore impossible when $D \leq 0$. Every positive interior optimum must have $D > 0$, which is equivalent to $\varepsilon(c_r^*; \alpha) < 1$.

Under the deployment-separable technology, the first-order condition becomes

$$G(c_r, \alpha) \equiv (B_s + w)D(c_r, \alpha) - A'(c_r) - P_c(c_r; \gamma) = 0.$$

Holding c_r fixed and differentiating the marginal deterrent return in α gives

$$D_\alpha(c_r, \alpha) = -\frac{e(c_r)\left[d_p(c_r; \alpha) - c_r \alpha e'(c_r)\right] + c_r e'(c_r) d_0}{d_p(c_r; \alpha)^3} < 0$$

whenever the emotional response is nonzero and $D > 0$. Hence $G_\alpha = (B_s + w)D_\alpha < 0$. By the maintained negative-curvature condition, $G_{c_r} = u_{cc} < 0$ at the interior optimum; the implicit function theorem therefore yields

$$\frac{\partial c_r^*}{\partial \alpha} = -\frac{G_\alpha}{G_{c_r}} < 0.$$

□

References cited in this appendix

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